

SAFETY DATA SHEET

SECTION 1: CHEMICAL PRODUCT AND IDENTIFICATION

PRODUCT NAME : Amoil Nautilus HLP Series
(10, 15, 22, 32, 46, 68, 100, 150, 220, 320)
PRODUCT TYPE : Premium Anti-Wear R&O Hydraulic Oils
SUPPLIER : Amoil (Malaysia) Sdn. Bhd.
ADDRESS : 14, Jalan Tembaga SD 5/2, Bandar Sri Damansara,
Kepong, 52200 Kuala Lumpur.
TEL : 603 – 62746563, 6273 4269
FAX : 603 – 6273 4267

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

<u>Composition</u>	<u>CAS No.</u>	<u>Wt%</u>
Mixture of highly-refined mineral oils and additives	Mixture	100

SECTION 3: HAZARDS IDENTIFICATION

HUMAN HEALTH HAZARDS:

No specific hazards under normal use conditions. Contains mineral oil for which an exposure limit for oil mist applies. Prolonged or repeated exposure may give rise to dermatitis. Used oil may contain harmful impurities.

SAFETY HAZARDS:

Not classified as flammable, but will burn.

ENVIRONMENTAL HAZARDS:

Not readily biodegradable. Expected to have a high potential to bioaccumulate.

OTHER INFORMATION:

Not classified as dangerous for supply or conveyance.

SECTION 4: FIRST AID MEASURES

SYMPTOMS AND EFFECTS:

Not expected to give rise to an acute hazard under normal conditions of use.

INHALATION:

In the unlikely event of dizziness or nausea, remove casualty to fresh air. If symptoms persist, obtain medical attention.

SKIN:

Remove contaminated clothing and wash affected skin with soap and water. If persistent irritation occurs, obtain medical attention. If high pressure injection injuries occur, obtain medical attention immediately.

EYE:

Flush eye with copious quantities of water. If persistent irritation occurs, obtain medical attention.

INGESTION:

Wash out mouth with water and obtain medical attention. Do not induce vomiting.

ADVICE TO PHYSICIANS:

Treat symptomatically. Aspiration into the lungs may result in chemical pneumonitis. Dermatitis may result from prolonged or repeated exposure.

SECTION 5: FIRE FIGHTING MEASURES
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SPECIFIC HAZARDS:

Combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide, oxides of sulphur, and unidentified organic and inorganic compounds.

EXTINGUISHING MEDIA:

Foam and dry chemical powder. Carbon dioxide, sand or earth may be used for small fires only.

UNSUITABLE EXTINGUISHING MEDIA:

Water in a jet. Use of Halon extinguishers should be avoided for environmental reasons.

PROTECTIVE EQUIPMENT:

Proper protective equipment including breathing apparatus must be worn when approaching a fire in a confined space.

SECTION 6: ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS:

Avoid contact with: skin and eyes.

PERSONAL PROTECTION: Wear impermeable gloves and boots.

ENVIRONMENTAL PRECAUTIONS:

Prevent from spreading or entering into drains, ditches or rivers by using sand, earth, or other appropriate barriers. Inform local authorities if this cannot be prevented.

CLEAN-UP METHODS-SMALL SPILLAGE:

Absorb liquid with sand or earth. Sweep up and remove to a suitable, clearly marked container for disposal in accordance with local regulations.

CLEAN-UP METHODS-LARGE SPILLAGE:

Prevent from spreading by making a barrier with sand, earth or other containment material. Reclaim liquid directly or in an absorbent. Dispose of as small spills.

SECTION 7: HANDLING AND STORAGE
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HANDLING:

When handling product in drums, safety footwear should be worn and proper handling equipment should be used. Prevent spillages.

STORAGE:

Keep in a cool, dry, well-ventilated place. Use properly labelled and closable containers. Avoid direct sunlight, heat sources, and strong oxidizing agents.

STORAGE TEMPERATURE: 0°C minimum to 50°C maximum.

RECOMMENDED MATERIALS:

For containers or container linings, use: mild steel or high density polyethylene.

UNSUITABLE MATERIALS: For containers or container linings, avoid: PVC.

OTHER INFORMATION:

Polyethylene containers should not be exposed high temperatures because of possible risk of distortion.

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

ENGINEERING CONTROL MEASURES:

Use local exhaust ventilation if there is a risk of inhalation of vapours, mists or aerosols.

OCCUPATIONAL EXPOSURE STANDARDS:

Threshold limit values are given below. Lower exposure limits may apply locally.

Component name	Limit type	Value	Unit	Other information
Oil mist, mineral	8 hour TWA	5	mg/m ³	ACGIH
	15min STEL	10	mg/m ³	ACGIH

HYGIENE MEASURES:

Wash hands before eating, drinking, smoking and using the toilet.

RESPIRATORY PROTECTION:

Not normally required. If oil mist cannot be controlled, a respirator fitted with an organic vapour cartridge combined with a particulate pre-filter should be used.

HAND PROTECTION: PVC or nitrile rubber gloves.

EYE PROTECTION:

Wear safety glasses or full face shield if splashes are likely to occur.

BODY PROTECTION:

Minimise all forms of skin contact. Wear overalls to minimise contamination of personal clothing. Launder overalls and undergarments regularly.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

AMOIL NAUTILUS HLP	10	15	22	32	46	68	100	150	220	320
PHYSICAL STATE:	Liquid at ambient temperature									
COLOUR:	Light brown									
ODOUR:	Characteristic mineral oil									

INITIAL BOILING POINT:	>280°C									
VAPOUR PRESSURE:	<0.5 Pa @ 20°C									
VISCOSITY, cSt @ 40°C	10	15	22	32	46	68	100	150	220	320
VAPOUR DENSITY (air=1):	>1 @ 20°C									
POUR POINT, °C:	-45	-45	-45	-45	-40	-40	-30	-12	-12	-12
FLASH POINT, °C :	190	190	190	200	200	210	210	230	230	240
FLAMMABILITY LIMIT – lower:	1% V/V (typical)									
FLAMMABILITY LIMIT – upper:	10% V/V (typical)									
AUTO-IGNITION TEMPERATURE:	>320° (typical)									
SOLUBILITY IN WATER:	Negligible									

The above data are typical values and do not constitute a specification. Vapor pressure data are calculated unless otherwise noted.

SECTION 10: STABILITY AND REACTIVITY

STABILITY: Stable

CONDITIONS TO AVOID: Extremes of temperature and direct sunlight.

MATERIALS TO AVOID: Strong oxidizing agents.

HAZARDOUS DECOMPOSITION PRODUCTS:

Hazardous decomposition products are not expected to form during normal storage.

SECTION 11: TOXICOLOGICAL INFORMATION

BASIS FOR ASSESSMENT:

Toxicological data have not been determined specifically for this product. Information given is based on knowledge of the components and the toxicology of similar products.

ACUTE TOXICITY – ORAL: LD50 expected to be above 2000 mg/kg.

ACUTE TOXICITY – DERMAL: LD50 expected to be above 2000 mg/kg.

ACUTE TOXICITY – INHALATION: Data not available.

EYE IRRITATION: Expected to be slightly irritant.

SKIN IRRITATION: Expected to be slightly irritant.

RESPIRATORY IRRITATION:

If mists are inhaled, slight irritation of the respiratory tract may occur.

SKIN SENSITIZATION: Not expected to be a skin sensitizer.

CARCINOGENICITY:

Product is based on mineral oils of types shown to be non-carcinogenic in animal skin-painting studies. Other components are not known to be associated with carcinogenic effects.

MUTAGENICITY: Not considered to be a mutagenic hazard.

OTHER INFORMATION:

Prolonged and/or repeated contact with this product can result in defatting of the skin, particularly at elevated temperatures. This can lead to irritation and possibly dermatitis, especially under conditions of poor personal hygiene. Skin contact should be minimized. Used oils may contain harmful impurities that have accumulated during use. The concentration of such impurities will depend on use and they may present risks to health and the environment on disposal. All used oil should be handled with caution and skin contact avoided as far as possible.

SECTION 12: ECOLOGICAL INFORMATION

BASIS FOR ASSESSMENT:

Ecotoxicological data have not been determined specifically for this product. Information given is based on a knowledge of the components and the ecotoxicology of similar products.

MOBILITY:

Liquid under most environmental conditions. Floats on water. If it enters soil, it will adsorb to soil particles and will not be mobile.

PERSISTENCE/DEGRADABILITY:

Not readily biodegradable. Major constituents are expected to be inherently biodegradable, but the product contains components that may persist in the environment.

BIOACCUMULATION: Has the potential to bioaccumulate.

ECOTOXICITY:

Poorly soluble mixture. Product is expected to be practically non-toxic to aquatic organisms, LC/EC50 > 100 mg/L. May cause physical fouling of aquatic organisms. (LC/EC50 expressed as the nominal amount of product required to prepare aqueous test extract)

SECTION 13: DISPOSAL CONSIDERATION

WASTE DISPOSAL:

Recycle or dispose of in accordance with prevailing regulations, preferably to a recognized collector or contractor. The competence of the contractor to deal satisfactorily with this type of product should be established beforehand.

SECTION 14: TRANSPORT INFORMATION

ICAO/ITATA	:	Not regulated.
IMDG (Ambient)	:	Not regulated.
IMDG (ELEVATED)	:	Do not transport – additional information required.
IMDG EMS	:	Not applicable.
IMDG MFAG	:	Not applicable.
IMO MARINE BULK	:	No not transport – additional information required.
USCG COMPATIBILITY	:	Not determined.

SECTION 15: REGULATORY INFORMATION

U.S. TSCA INVENTORY:

All components of this material are on the US TSCA Inventory or are exempt.

OTHER TSCA REG.: None known.

SARA EXT. HAZ. SUBST.:

This product does not contain greater than 1.0% of any chemical substance on the SARA Extremely Hazardous Substances list.

SARA SECTION 313:

This product does not contain greater than 1.0% (greater than 0.1% for carcinogenic substance) of any chemical substances listed under SARA Section 313.

TDG REGULATED LIMIT: None known.

CERCLA HAZARDOUS SUBSTANCES: None known.

CAL. PROP. 65:

This product does not intentionally contain any chemicals known by the State of California to cause cancer and/or birth defects. Moreover, AMOIL does not routinely analyze its products for impurities which may be such chemicals.

U.S. FUEL REGISTRATION: Not applicable.

U.S. DEPT. OF AGRICULTURE:

This product has not been filed with the USDA to support H2 approvals.

NSF NONFOOD COMPOUNDS REGISTRATION:

This product has not been filed with NSF to support H1 or H2 approvals.

FDA APPROVAL: Not applicable.

SECTION 16: OTHER INFORMATION

The information presented herein has been compiled from sources considered to be dependable and is accurate to the best of The Amoil Company's knowledge; however, The Amoil Company makes no warranty whatsoever, expressed or implied of MERCHANTABILITY OR FITNESS FOR THE PARTICULAR PURPOSE, regarding the accuracy of such data or the result to be obtained from use thereof. The Amoil Company assumes no responsibility for injury to recipient or to third persons or for any damage to any property and recipient assumes all such risks.

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